

CE 310 — Week 3 Lab: Condition or Miss the Point

Dataset: DOE Reference Building — Medium Office, ASHRAE Climate Zone 2B (Tucson Area), 2023 Topic: Marginal Probability · Conditional Probability · Joint Probability · Independence

Before You Begin

Files needed: - Week3_Tucson_Office_Energy_2023.csv — hourly building energy data (8,760 rows) - Week3_Answer_Sheet.xlsx — download from D2L and enter all answers here

Dataset note: This is real simulation output from the U.S. Department of Energy's Reference Building program — a medium-sized office building (3 floors, ~53,600 sq ft) calibrated to Tucson's climate. Each row is one hour of 2023.

Answer Sheet instructions: Type every numerical answer in Col E of the row matching the question label — A1 in row A1, B3 in row B3. For ★ questions, also enter the intermediate value in Col D. For Col C, type the Excel formula you used.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Background

Today you'll move from $P(\text{event})$ to $P(\text{event} \mid \text{condition})$ — and see how much a condition can change the picture.

Excel Quick Reference

Goal	Formula / Method
Count non-empty cells in a column	=COUNTA(A:A)-1 — subtract 1 to exclude the header row
Count hours meeting one condition	=COUNTIF(range, ">50")
Count hours meeting two conditions	=COUNTIFS(range1, ">50", range2, "Weekday")
Average for one group	=AVERAGEIF(B2:B8761, 7, G2:G8761) — average Elec where Month=7

Goal	Formula / Method
Average for two conditions	=AVERAGEIFS(H2:H8761, E2:E8761, "Summer", F2:F8761, "Yes")
Binary indicator (0/1)	=IF(H2>50, 1, 0) — fills down for pivot table use
Insert pivot table	Insert tab → PivotTable → configure Rows, Values
Average in pivot table	Values area → Value Field Settings → Average
Conditional probability	= joint count / group total × 100
Independence test	$P(A \cap B)$ vs $P(A) \times P(B)$ — compute both and compare

Setup (~10 min)

1. Import Week3_Tucson_Office_Energy_2023.csv into a new Excel workbook:
 - Data → Get Data → From Text/CSV → Load.
2. Confirm 8,760 rows of data (use =COUNTA(A:A)-1 to verify).
3. Confirm the 12 column names in row 1:
 - DateTime, Month, Hour, DayType, Season, BusinessHours
 - Electricity_kWh, Cooling_kWh, HeatingElec_kWh, Lighting_kWh, Equipment_kWh, Gas_kWh
 - DayType = "Weekday" or "Weekend" · Season = "Winter/Spring/Summer/Fall"
 - BusinessHours = "Yes" (Mon–Fri 8am–6pm) or "No"
4. Add two helper columns — you will use these for the pivot table in Section B:
 - In Column M, header IsCoolHigh:
=IF(H2>50, 1, 0)
Fill down to row 8761.
 - In Column N, header IsElecHigh:
=IF(G2>100, 1, 0)
Fill down to row 8761.

Why 0/1? A binary indicator column turns a condition into a number. The average of a 0/1 column equals the fraction of rows where the condition is true — which is a probability. When you pivot on it, Excel computes conditional probabilities automatically.

Section A — Marginal Probabilities (~25 min)

Marginal probability = probability of an event without conditioning on anything else.

All Section A questions use COUNTIF or COUNTIFS on a single column. Round percentages and kWh values to 2 decimal places.

A1 — Hours With Any Cooling Load

How many hours in 2023 did the building use any cooling energy (Cooling_kWh > 0)?

This tells you how many hours the cooling system ran at all — including partial loads too small to matter for design.

A2 — Hours With High Cooling Demand

How many hours had Cooling_kWh > 50 — the threshold where the system is operating at a substantial fraction of design capacity?

A3 — Marginal P(Cooling > 50)

Compute the marginal probability that any randomly chosen hour has high cooling demand:

$$P(\text{Cooling} > 50) = \text{A2 value} / 8760 \times 100$$

Express as a percentage, rounded to 2 decimal places.

This is the marginal probability. Section B will show that it dramatically understates the risk during peak conditions — and overstates it during winter.

A4 — Hours With High Total Electricity

How many hours had Electricity_kWh > 100 — above the building's nominal base load?

A5 — Marginal P(Electricity > 100)

$$P(\text{Electricity} > 100) = \text{A4 value} / 8760 \times 100$$

A6 — Count of Summer Hours

How many of the 8,760 hours fall in Summer (Season = "Summer")?

A7 — Count of Weekday Hours

How many hours are on weekdays (DayType = "Weekday")?

A8 — July Average Electricity

Calculate the average hourly electricity demand for July only (Month = 7):

Compare this to the full-year mean (~87.4 kWh). July's higher average hints at what conditioning reveals in Section B.

Section B — Conditional Probabilities

Report percentages to 2 decimal places, kWh values to 2 decimal places.

B1 — Joint Count: High Cooling AND July

How many hours had both Cooling_kWh > 50 and Month = 7 (July)?

B2 ★ — $P(\text{Cooling} > 50 \mid \text{July})$

The conditional probability of high cooling demand given that the month is July:

$$P(\text{Cooling} > 50 \mid \text{July}) = (\text{hours with Cooling} > 50 \text{ AND Month} = 7) / (\text{total July hours}) \times 100$$

Col D: total July hours (=COUNTIF(B2:B8761, 7)) · Col E: $P(\text{Cooling} > 50 \mid \text{July})$ as %

Compare this to your A3 marginal value. The conditioning effect — the ratio $P(\text{condition}|\text{context}) \div P(\text{condition marginal})$ — shows how much the month changes the probability.

B3 — $P(\text{Cooling} > 50 \mid \text{January})$

Repeat for January (Month = 1):

Tucson winters are mild, but not warm enough for cooling. Does a zero probability here mean the building uses no energy in January? (Check column G for January rows — January electricity averages about 82 kWh/hr and stays above 30 kWh/hr in roughly 92% of hours, but it comes from lighting and plug loads, not cooling.)

Complement rule: $P(\text{Cooling} \leq 50 \mid \text{January}) = 1 - \text{B3}$. No computation needed — state the value.

B4 — Independence Test: Cooling Demand and Month

If high cooling demand were statistically independent of month, then:

$$P(\text{Cooling} > 50 \text{ AND July}) \text{ would equal } P(\text{Cooling} > 50) \times P(\text{July})$$

Compute this independence product:

$$= (A3 \text{ value} / 100) \times (\text{July_hours} / 8760) \times 100$$

where July_hours = 744 (the B2 intermediate value — do not use cell A6, which counts Summer hours = 2,208).

Express as a percentage.

If this independence product matches the actual joint probability $P(C > 50 \text{ AND July}) = B1/8760 \times 100$, the variables are independent. If not, conditioning on month changes the probability — and the variables are associated.

B5 — Joint Count: High Electricity AND Weekday

How many hours had both Electricity_kWh > 100 and DayType = “Weekday”?

B6 ★ — $P(\text{Electricity} > 100 \mid \text{Weekday})$

$$P(E > 100 \mid \text{Weekday}) = B5 \text{ value} / \text{total weekday hours} \times 100$$

Col D: total weekday hours (your A7) · Col E: $P(E > 100 \mid \text{Weekday})$ as %

Engineering Context: NFPA 110 requires backup generators sized for the maximum anticipated connected load — not the average load.

B7 — $P(\text{Electricity} > 100 \mid \text{Weekend})$

Compare B6 to B7. The ratio $P(E > 100 \mid \text{Weekday}) / P(E > 100 \mid \text{Weekend})$ is the lift — a metric used in data science to measure how much a condition changes a probability. A lift > 1 means positive association; lift < 1 means negative association.

B8 — Independence Product: Electricity and Day Type

Compute the independence product — what $P(\text{Electricity} > 100 \text{ AND Weekday})$ would be if the two events were unrelated:

$$\text{Independence product} = (A5 \text{ value} / 100) \times (A7 \text{ value} / 8760) \times 100$$

Then compute the actual joint probability for comparison (do NOT enter this in the answer cell):

$$\text{Actual joint } P(E > 100 \text{ AND Weekday}) = B5 \text{ value} / 8760 \times 100$$

If the independence product equals the actual joint probability, the events are independent. If they differ, they are associated. Which is larger here, and

what does that tell you about whether high electricity demand is more or less common on weekdays than chance would predict?

Addition rule check (ungraded): Compute $P(\text{Electricity} > 100 \text{ OR Weekday}) = P(E > 100) + P(\text{Weekday}) - P(E > 100 \text{ AND Weekday})$, where $P(\text{Weekday}) = A7/8760 \times 100$ and $P(E > 100 \text{ AND Weekday}) = B5/8760 \times 100$. Show the arithmetic.

B9 ★ — $P(\text{Electricity} > 100 \mid \text{Business Hours})$

How likely is high electricity demand during official business hours (Mon–Fri 8am–6pm)?

Col D: business-hours count ($=\text{COUNTIF}(F2:F8761, "Yes")$) · Col E: $P(E > 100 \mid \text{BusinessHours} = \text{Yes})$ as %

B10 — $P(\text{Electricity} > 100 \mid \text{NOT Business Hours})$

The contrast between B9 and B10 is the central result of this lab. An engineer or facility manager who sizes standby systems based on the marginal 39.3% probability (A5) would both over-provision off-hours capacity and under-provision business-hours capacity. Conditioning on schedule resolves both problems simultaneously.

ArcE Framing: ASHRAE 90.1 requires HVAC systems sized for design conditions, not average conditions. A system sized for the annual average cooling demand (18.96 kWh/hr) would be severely undersized during summer business hours.

B11 — Pivot Table: $P(\text{Cooling} > 50 \mid \text{Season})$

Use the IsCoolHigh helper column (Col M) to build a pivot table:

Step 1 — Insert Pivot Table: 1. Click any cell in your data. Insert tab → PivotTable → choose “New Worksheet” → OK.

Step 2 — Configure fields: - Drag Season to the Rows area. - Drag IsCoolHigh to the Values area. - In the Values area, click the dropdown → Value Field Settings → change from “Sum” to “Average”.

Step 3 — Read the result: The pivot table now shows the average of IsCoolHigh for each season. Since IsCoolHigh is 0 or 1, the average equals the fraction of hours with Cooling > 50 — which is $P(\text{Cooling} > 50 \mid \text{Season})$.

Multiply each value by 100 to convert to percentage. Enter the Summer value as your answer.

Alternative (no Pivot Table required): If your version of Excel does not support Pivot Tables, use COUNTIFS to divide the count of Summer hours with Cooling

> 50 kWh by the total count of Summer hours. The result is identical to the pivot table average.

B12 — Month With Highest Conditional Cooling Probability

Extend your pivot table to show $P(\text{Cooling} > 50 \mid \text{Month})$: - Remove Season from the Rows area, drag Month in its place. - The table now shows $P(\text{Cooling} > 50)$ for each of the 12 months.

Which month (enter the month number) has the highest conditional probability of high cooling demand?

Alternative (no Pivot Table required): In a blank area of the sheet, build a small two-column table with months 1–12 in one column. In the adjacent column, compute $P(\text{Cooling} > 50 \mid \text{Month} = m)$ for each row using COUNTIFS — dividing the count of hours in that month with Cooling > 50 kWh by the total count of hours in that month, multiplied by 100. Replace m with the month number (1 = Jan, 6 = Jun, 7 = Jul, etc.). Identify the row with the maximum value — that month number is your answer.

B13 — Mean Cooling During Peak Conditions

Calculate the average cooling demand during the highest-load hours: Summer season and business hours:

Compare to the annual mean cooling (~19 kWh/hr). The ratio shows how much the peak-condition mean exceeds the marginal mean — this is the practical consequence of using $P(\text{event})$ instead of $P(\text{event} \mid \text{design condition})$ for system sizing.

B14 ★ — Probability Bridge: $P(\text{Weekday} \mid \text{July})$

So far you have computed $P(\text{event} \mid \text{time condition})$. Now reverse the conditioning:

$P(\text{Weekday} \mid \text{July}) = ?$

Col D: count of Weekday AND July hours · Col E: ÷ total July hours (from B2) × 100

Notice: $P(\text{Weekday} \mid \text{July}) \neq P(\text{Weekday})$ overall. July has the same weekday/weekend structure as every other month — so why do they differ? They shouldn't differ by much if the calendar is symmetric. In this week's lecture you learned Bayes' theorem for reversing conditional probabilities systematically: $P(B|A) = P(A|B) \cdot P(B) / P(A)$. You can verify your B14 answer two ways — directly by COUNTIFS and by applying Bayes to your B2 result — and both should match. This is the foundation of probabilistic reasoning and of the Naive Bayes classifier.

Target shape for Chart 1

Figure 1: Target shape for Chart 1

Target shape for Chart 2

Figure 2: Target shape for Chart 2

Required Charts

Insert both charts in your working Excel file.

Chart 1 — Monthly Conditional Probability Profile (7 pts)

From your pivot table (Month as rows, Average of IsCoolHigh as values), build a bar chart. Multiply y-values by 100 to show percentages. Add a horizontal reference line at your A3 marginal value.

- Title: P(Cooling > 50 kWh | Month) — Tucson Office Building 2023
- Axes: Month (1–12) on x, Conditional Probability (%) on y

Chart 2 — Conditioning Effect Comparison (8 pts)

Build a 5-row table with these conditions and their P(Electricity > 100 kWh) values: Marginal (all hours), Weekday, Weekend, Business Hours, Non-Business Hours. Plot as a clustered bar chart with data labels on each bar.

- Title: How Conditioning Changes P(Electricity > 100 kWh) — Tucson Office 2023
 - Axes: condition labels on x, Probability (%) on y
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Memo Deliverable

Type your memo directly in the Written sheet of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed.

To: CE 310 Instructor

From: [Your Name], [Your Major]

Date: [Date]

Subject: Week 3 Lab — Conditional Probability Analysis, Tucson Office Building 2023

Paragraph 1 — Marginal vs. Conditional: State the marginal P(Cooling > 50 kWh) and P(Electricity > 100 kWh). Then describe how conditioning on month or time of day changes these probabilities. Cite at least two specific conditional probability values from your Section B results. Explain in one sentence why the marginal probability is misleading for design purposes.

Paragraph 2 — The Conditioning Effect: Using your Chart 2, describe the range of $P(\text{Electricity} > 100 \text{ kWh})$ across your five conditions. What does the ratio $P(E > 100 | \text{BusinessHours}) / P(E > 100 | \text{NonBusinessHours})$ tell you about the relationship between occupancy schedule and electricity demand? Reference the independence test result from B4 or B8.

Paragraph 3 — Cooling system sizing and conditional probability:

An HVAC engineer is sizing a chilled-water cooling system for a medium office building in Tucson. Using your B11, B12, and B13 results, explain why the ASHRAE 90.1 design conditions require conditioning on season and time of day, not just annual average cooling load. Cite at least one specific conditional probability from your pivot table.

Submission: Save as CE310_W3_<NetID>.xlsx and upload to the Week 3 D2L dropbox. A 5-point deduction applies if your file cannot be matched to your student record.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
 - ☐ File saved as .xlsx and named CE310_W3_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in Col D
 - ☐ Uploaded to D2L before the deadline
-

Grading Summary

Component	Points
Section A — A1–A8	16
Section B — B1–B14	60
Chart 1 — Monthly conditional probability profile	7
Chart 2 — Conditioning effect comparison	8
Memo (3 paragraphs)	9
Total	100